



Dimensionless Controls of Plasticity Under Alternating Tasks: From Evolutionary Biology to Continual Learning

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Introduction

Plasticity (sustained performance as conditions change) is studied in parallel in continual learning and evolutionary biology. García-Galindo and Ahnert [1] show that biological plasticity emerges when environments alternate, and identify four controlling factors: **neutral set size, Hamming distance, switching period, and mutation rate**. We translate their setting to a deep network trained alternately on two label sets and ask which of these controls survive the passage to gradient descent.

We follow techniques from the emerging mechanics of learning [2] and identify two dimensionless controls: (1) The **task disagreement** $r = d_H/N \in [0, 1]$ is the fraction of inputs on which y_1 and y_2 differ. (2) The **reach** ηT is the product of learning rate and switching period: how far the network travels in parameter space during one phase. With both theory and computational experiments, we provide a new solvable toy model of continual learning with scaling laws, bounds, and dimensionless controls.

Experimental Setup

A network is trained alternately on two 7-bit Boolean label sets y_1 and y_2 over all $N = 2^7 = 128$ inputs, switching every T steps. After pre-training on y_1 , we alternate SGD between label sets until steady state. Sweeping 27 function pairs spanning low, medium, and high neutral set size, complexity, and Hamming distance across 6 learning rates $\times$ 6 switching periods $\times$ 10 trials gives 9,720 trajectories.

Accuracy Proxies: Losses are mapped to the unit square and called accuracies via the monotone transformation $(\ell_1, \ell_2) \mapsto \mathbf{a} = (e^{-\ell_1}, e^{-\ell_2})$.

Quantifying Plasticity: The **utopia distance** $d_U = \mathbb{E}_t \|\mathbf{a} - \mathbf{1}\|_2$ is the steady-state expected distance from the accuracy to the point (1,1), so lower utopia distance corresponds to higher plasticity. The **forgetting amplitude** measures the peak-to-peak swing in task 1 performance and is defined as $\Delta a_1 = \sup_{t,s \in [0, 2T]} |a_1(t) - a_1(s)|$.

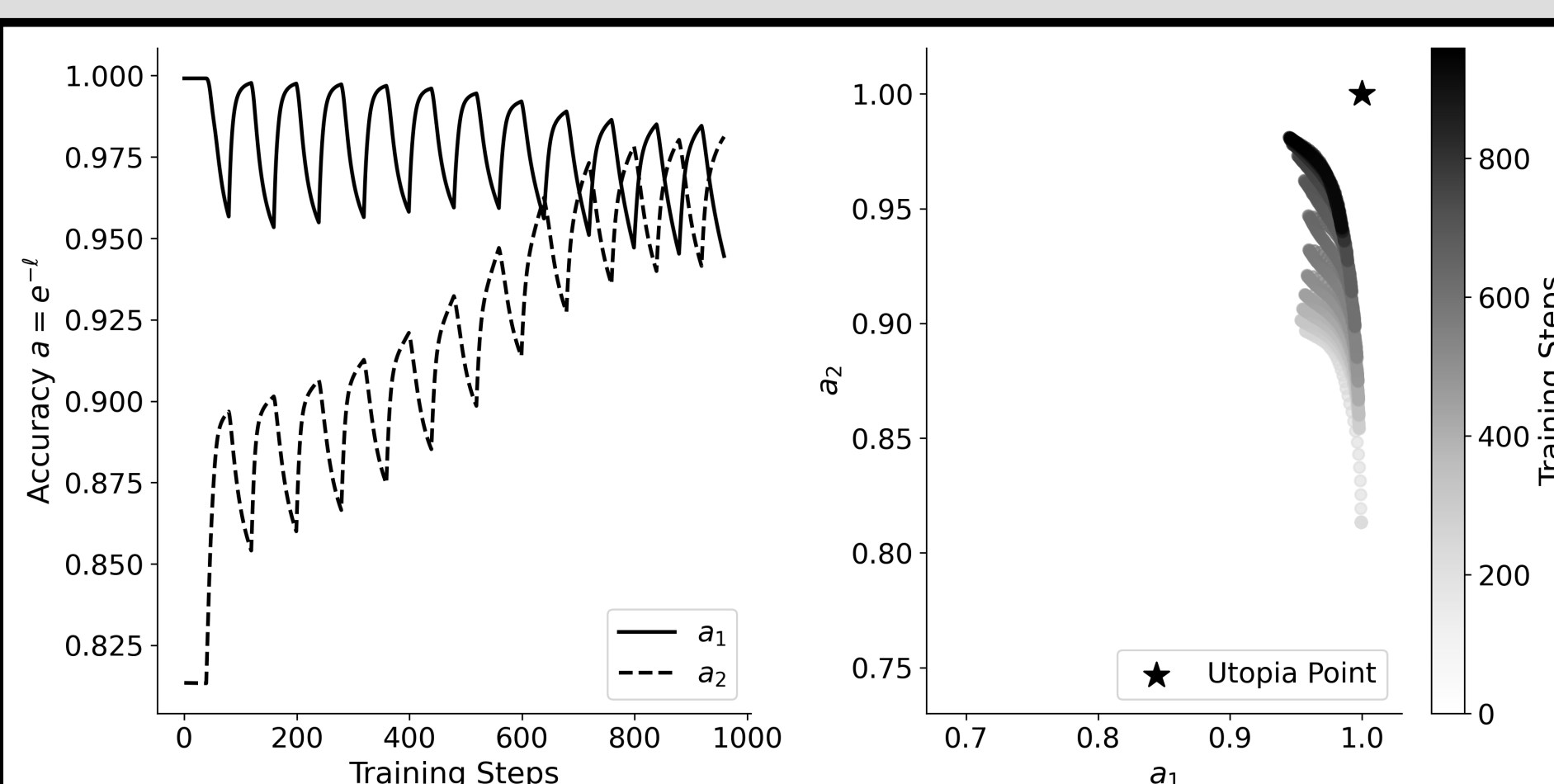


Figure 1. Example trajectory of alternating training on two functions. The left panel shows the accuracy proxies $a_1(t)$ and $a_2(t)$ over iterations. The right panel shows the corresponding phase plane trajectory and utopia point over iterations.

Theory

Theorem 1. Suppose the trajectory has reached a period- $2T$ steady-state orbit, so that $a_1(0) = a_1(2T)$; that $a_1(t)$ monotonically decreases during the y_2 training phase; and that the loss is L -smooth and the network f is C -smooth. Let $M = \sup_{\theta \in \text{Orbit}} \|\nabla \ell_1(\theta)\|_2 \max_{i \in \perp} \|J_i(\theta)\|_2 \leq LC < \infty$ where $\perp$ denotes the set of indices where the labels y_1 and y_2 differ, and $J_i(\theta)$ is the Jacobian of the network output w.r.t parameters θ at index i . Then $\Delta a_1 \leq c_1 = a_1(0)(e^{r\eta TM + O(\eta^2 T)} - 1)$. Further, for sufficiently small η , we have $c_1 \approx a_1(0)(e^{r\eta TM} - 1)$.

Theorem 2. If $0 < r < 1$ then $d_U \geq \sqrt{2}(1 - 2^{-r})$.

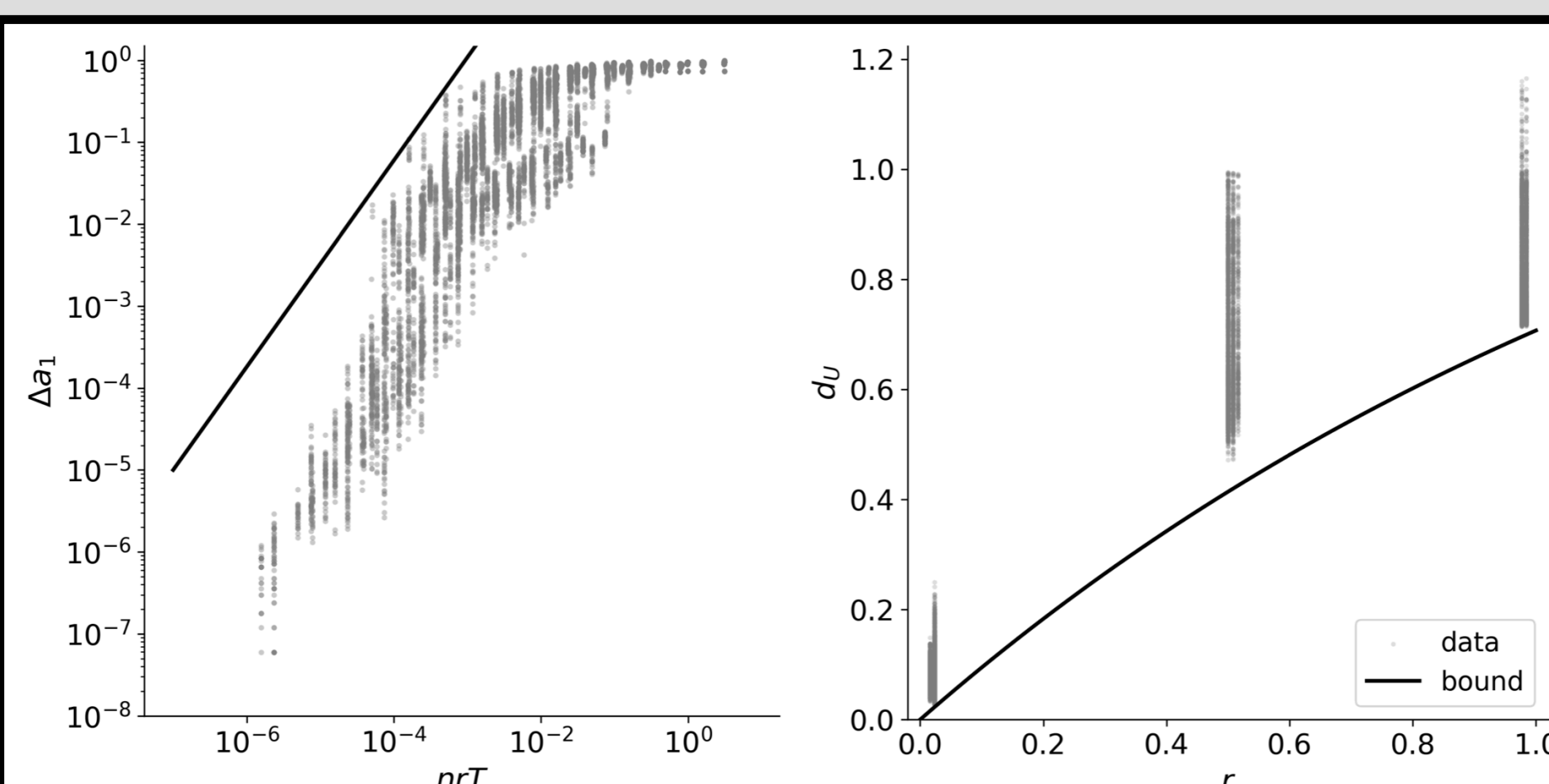


Figure 2. Empirical demonstration of the bounds of Theorems 1 (left) and 2 (right) across the 9,720 trajectories. For Theorem 1, we use the uniform estimate $M = 100$.

Identifying Controls

Table 1. Type II ANOVA effect sizes for the two log-transformed plasticity outcomes. Models use complexity bin, neutral-set-size class, Hamming-distance class, learning rate, and switching period as categorical predictors. Reported values are ω^2 effect sizes. Significance levels are denoted by *** for $p < 0.001$.

Factor	$\log \Delta a_1$	$\log d_U$
$\tilde{K}$ bin	1×10^{-4}	3×10^{-4} ***
NSS (ν) class	3×10^{-4} ***	1×10^{-4} ***
Hamming (r) class	0.4966 ***	0.9422 ***
η	0.2178 ***	0.0019 ***
T	0.1348 ***	0.0030 ***
Residual	0.1503	0.0525
R^2	0.8497	0.9475

The ANOVA confirms that the quantities showing up in the two theorems dominate (the learning rate, switching period, and task disagreement).

The ANOVA rejects the hypothesis that neutral set size and complexity strongly influence plasticity as they do in biology. This points to a structural difference between evolution on the fitness landscape of genotype-phenotype maps and learning on the loss landscape of neural networks.

Role of the Reach Parameter

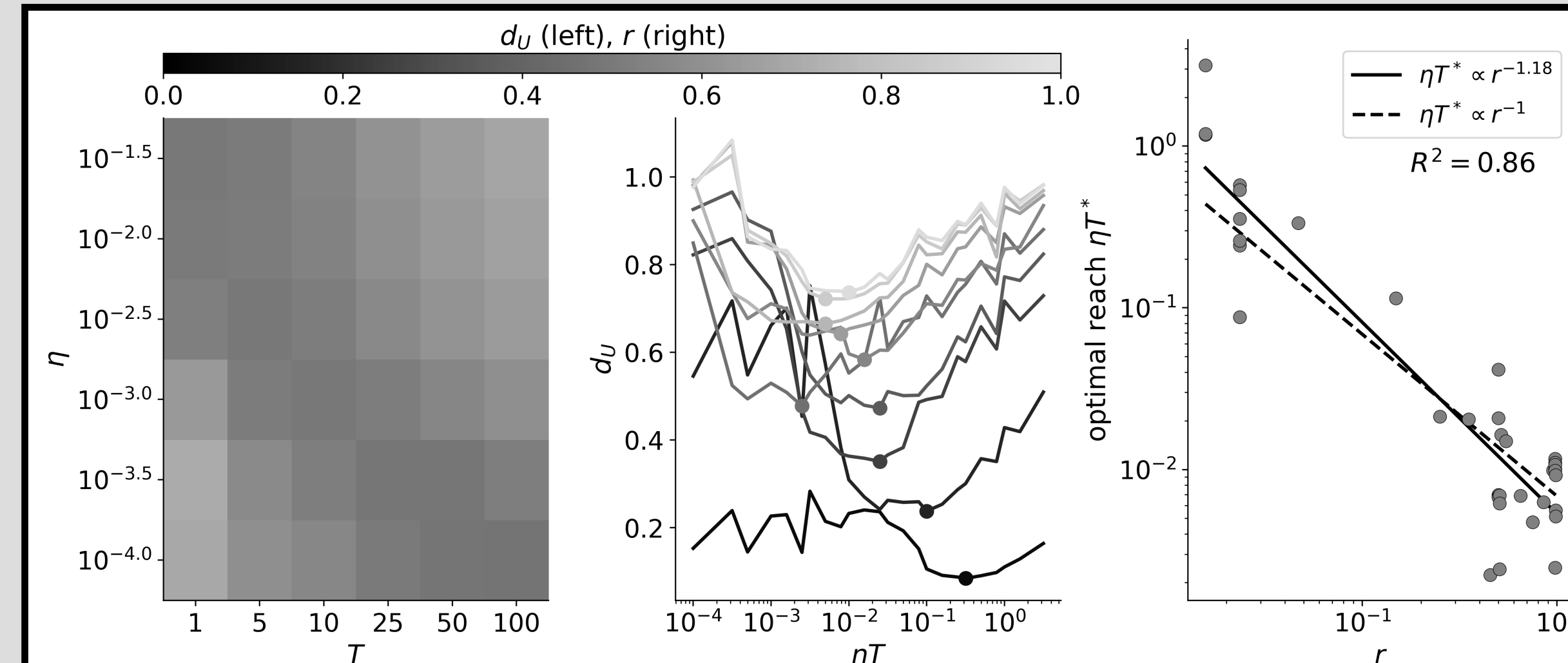


Figure 3. Heatmap of d_U values across learning rate/switching period space with the most plastic trajectories (lowest utopia distance values) occurring along a diagonal (constant reach) (left). Task disagreement r level curves of the utopia distance versus nominal reach (center) show that as r grows, the optimum rises and shifts to smaller reaches while curves become more convex. The optimal reach versus r over 37 task pairs (right) shows an approximate power-law fit.

The reach is approximately constant across the diagonal of optimal plasticity conditions (lowest utopia distance). As r increases, the minimum attainable utopia distance increases, but the optimal reach falls and sharpens as the r -level curves grow more convex. The optimal reach is approximately inversely proportional to the task disagreement r (true exponent -1.18 ± 0.17).

The reach-governed picture is not an artifact of the setup. It persists under minibatch SGD, momentum, and Adam, and under held-out continuous-input tasks measured on unseen test data (mean train-test gap in d_U below 0.01). Varying width and depth over a $16\times$ range in parameter count leaves d_U essentially unchanged, giving direct support for network-size independence.

Discussion

Task disagreement r alone sets a geometric floor on the utopia distance, while the reach ηT governs how far a trajectory sits above that floor. Their product ηT sets the scale of forgetting. Because $\eta T^* \approx 1/r$, the optimal reach can be determined solely from r , without probing the loss landscape or architecture. In other fields, quantities like ηT are Damköler numbers, or ratios between “reactivity” and “residence time” of a system. Here, $\eta T \ll 1$ so the system is reaction-limited, and optimal plasticity requires phases too short for relaxation to complete.

Future Work

The framework generalizes along three axes: (1) replacing the Hamming-based r with continuous dissimilarity measures for soft labels; (2) linking the switching period to the non-commutativity of the two gradient flows via the Lie bracket $[\nabla \ell_1, \nabla \ell_2]$, which would give the reach a geometric reading as curvature-weighted distance; and (3) testing whether neutral set size (negligible for a single trajectory) regains influence under ensembles or explicit noise injection, which sample the neutral set in a way more akin to how populations do in evolutionary biology.

Key References

- [1] Paula García-Galindo and Sebastian E. Ahnert. RNA plasticity emerges as an evolutionary response to fluctuating environments. *bioRxiv*, 2025. ISSN 2692-8205. doi: 10.1101/2024.10.02.614758. URL <https://www.biorxiv.org/content/10.1101/2024.10.02.614758v2>.
- [2] Jamie Simon, Daniel Kunin, Alexander Atanasov, Enric Boix-Adserà, Blake Bordelon, Jeremy Cohen, Nikhil Ghosh, Florentin Guth, Arthur Jacot, Mason Kamb, Dhruva Karkada, Eric J. Michaud, Berkan Özlük, and Joseph Turnbull. There will be a scientific theory of deep learning, 2026. URL <https://arxiv.org/abs/2604.21691>.

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